

Bagging vs Boosting (Ensemble Learning)

◆ Introduction

Bagging and Boosting are **ensemble learning techniques** in machine learning. They combine multiple weak models to create a stronger and more accurate model.

Both aim to improve performance, but they work in different ways.

1. Bagging (Bootstrap Aggregation)

Definition

Bagging is a method where multiple models are trained **independently and in parallel** on different random samples of data.

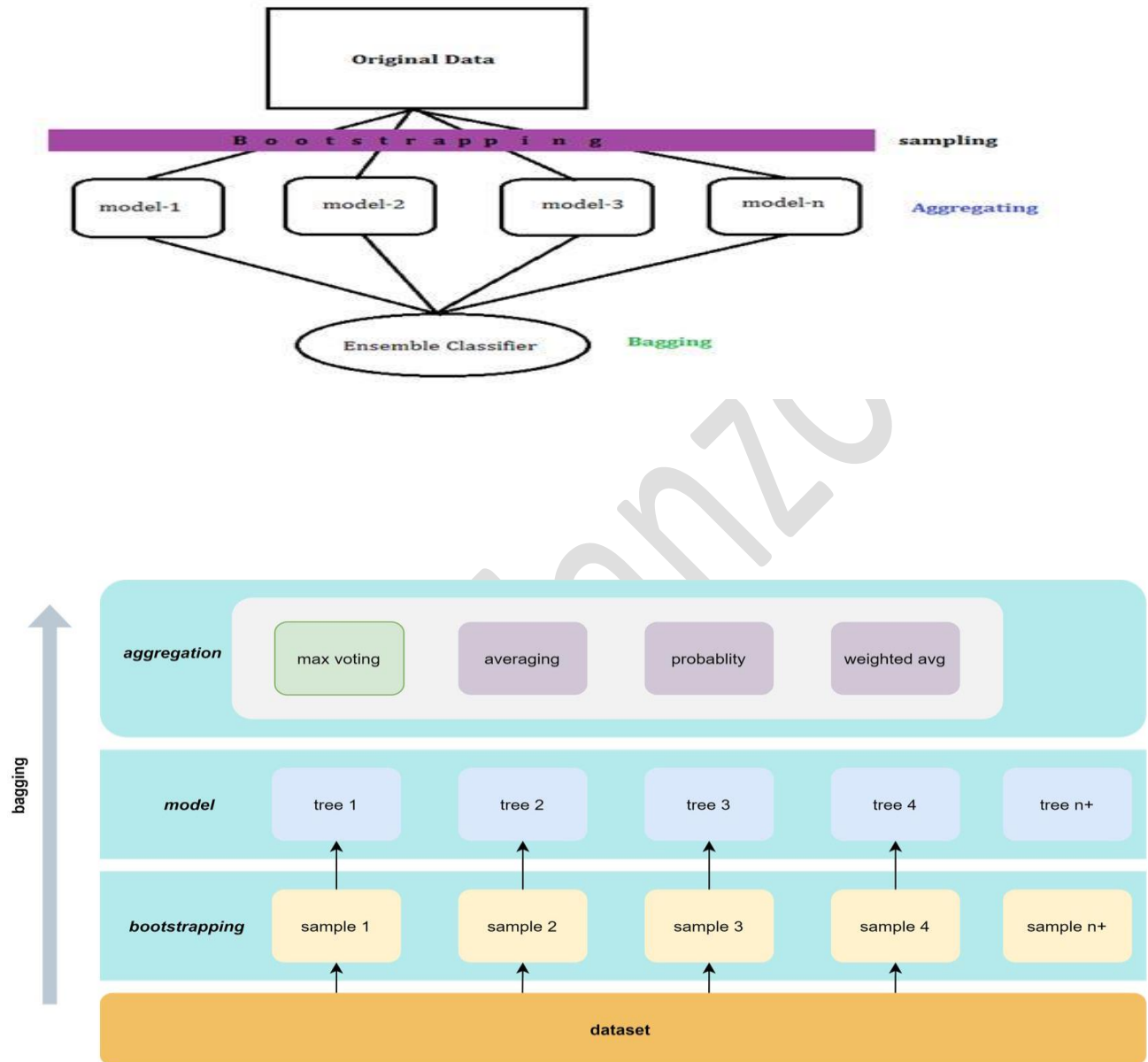
How it works

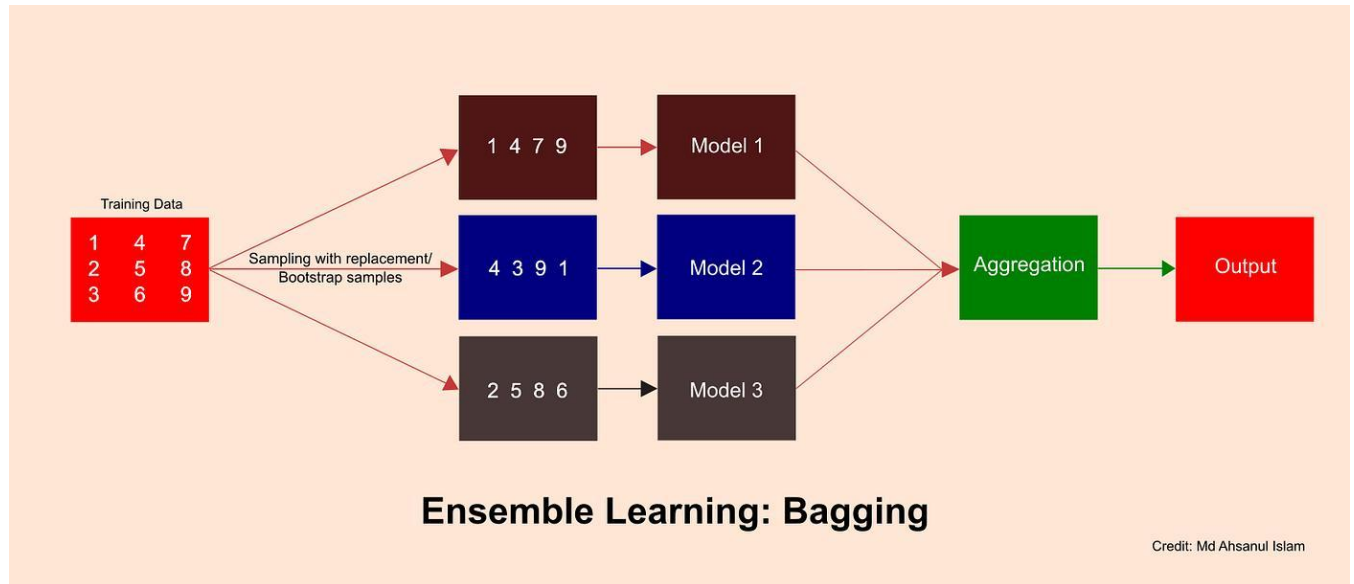
- Dataset is split into multiple random samples
- Each sample trains a separate model
- All models work independently
- Final output is combined using voting or averaging

Goal

- Reduce **variance**
- Prevent overfitting
- Improve stability

Visual Representation





Example Algorithm

- Random Forest

2. Boosting

Definition

Boosting is a method where models are trained **sequentially**, and each new model learns from the mistakes of the previous one.

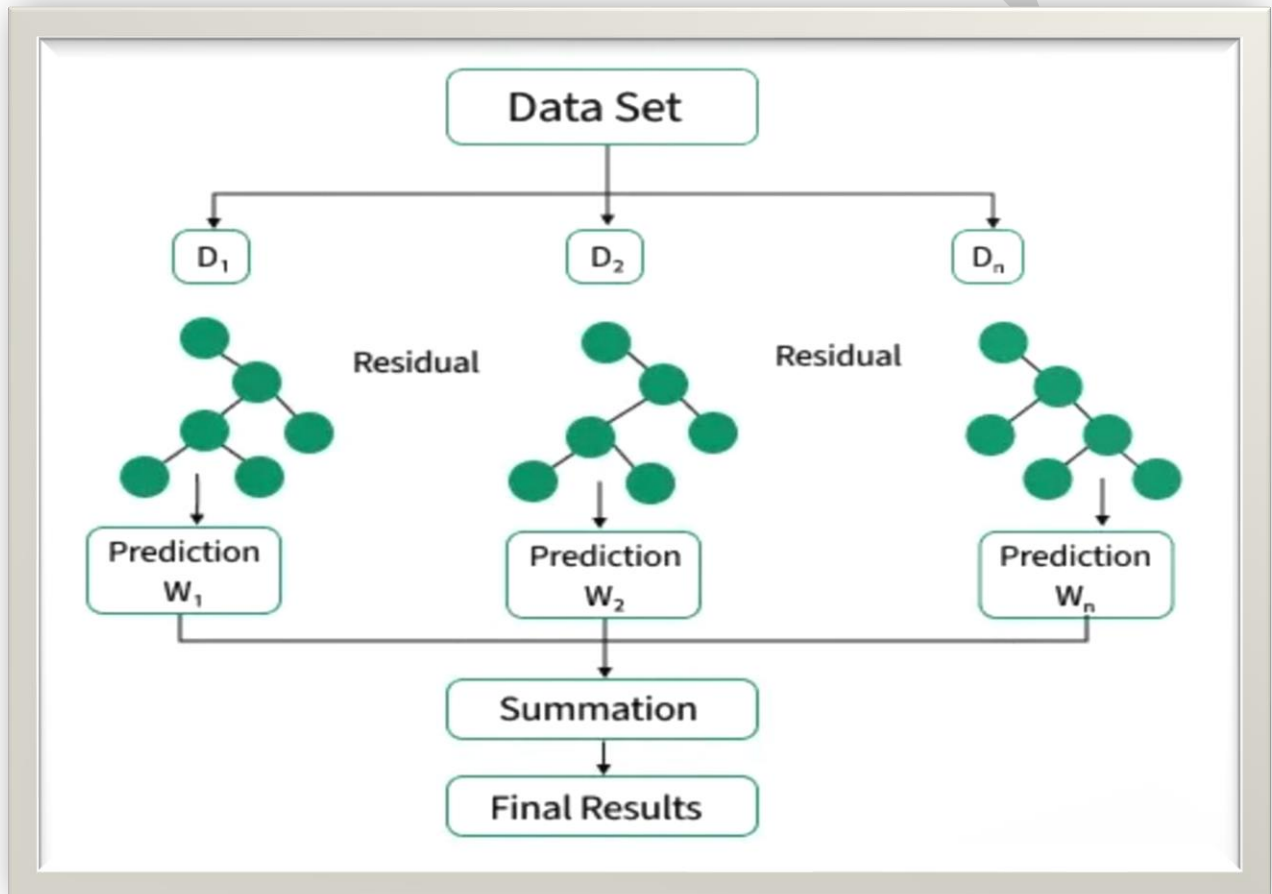
How it works

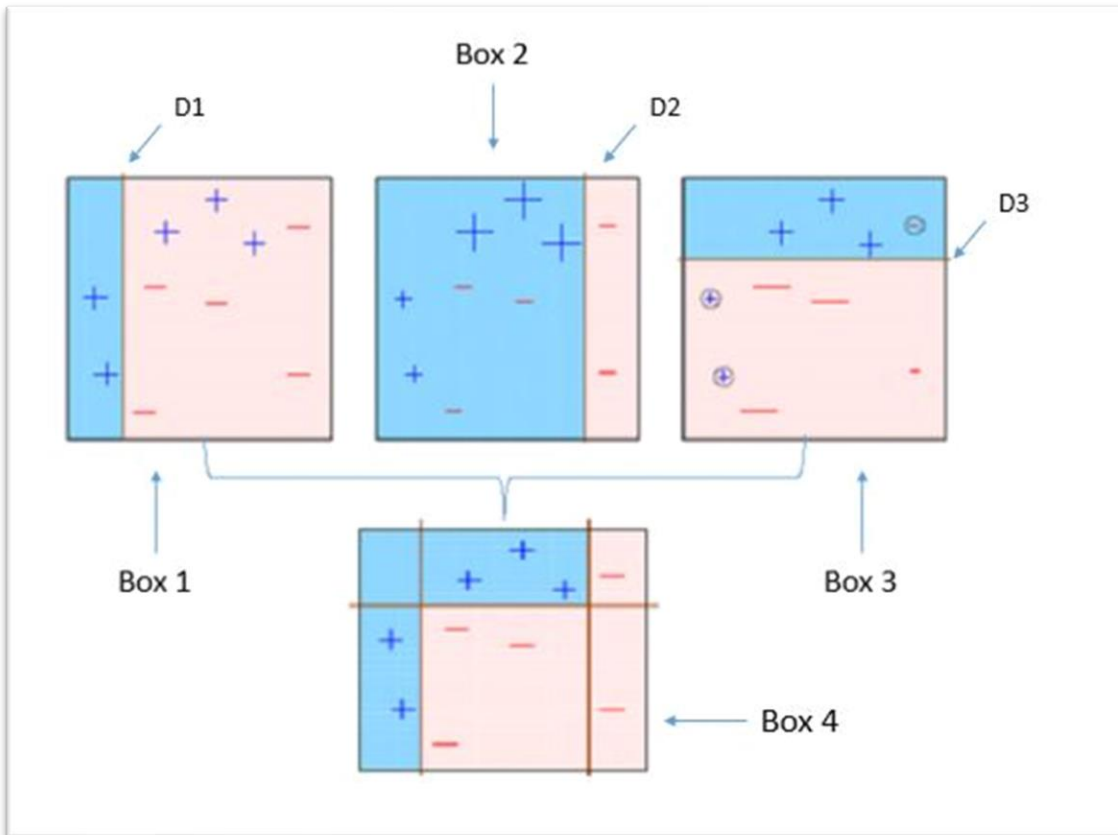
- First model is trained on full dataset
- Errors are identified
- Next model focuses more on wrong predictions
- Process continues step by step
- Final result is weighted combination of all models

Goal

- Reduce **bias**
- Improve accuracy
- Focus on difficult samples

Visual Representation





Gradient Boosted Decision Trees

$$\hat{y}_i^1 = f_1(x_i)$$



$$f_1(x_i) \rightarrow y_i$$

$$\hat{y}_i^2 = \hat{y}_i^1 + f_2(x_i)$$



$$f_2(x_i) \rightarrow y_i - \hat{y}_i^1$$

$$\hat{y}_i^M = \hat{y}_i^{M-1} + f_M(x_i)$$



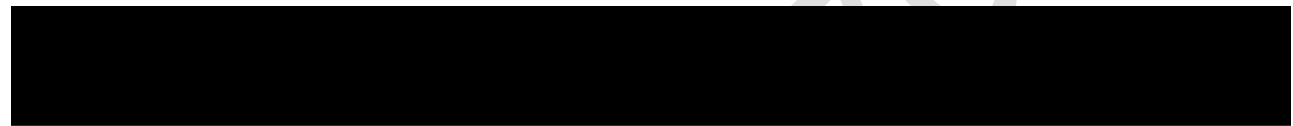
$$f_M(x_i) \rightarrow y_i - \hat{y}_i^{M-1}$$

Example Algorithms

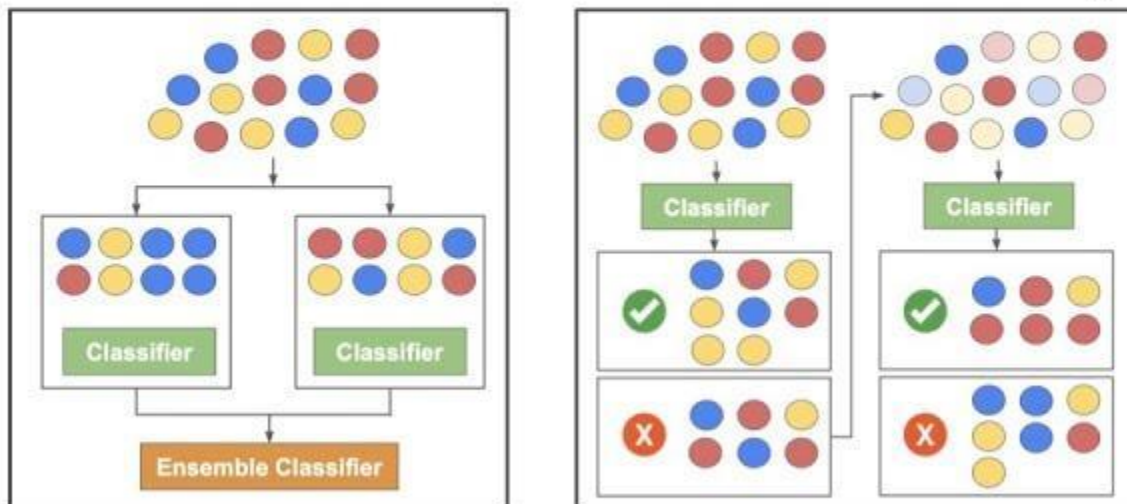
- AdaBoost
- Gradient Boosting
- XGBoost

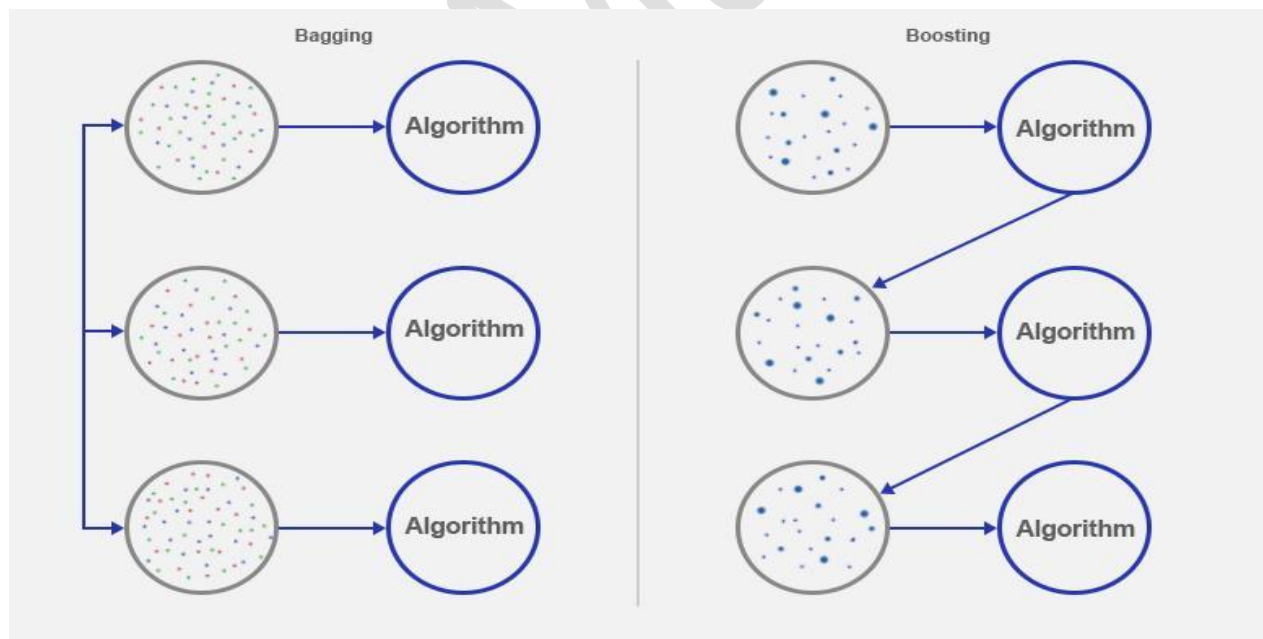
3. Bagging vs Boosting (Main Differences)

Simple Comparison Diagram



Bagging vs Boosting





Key Differences

Feature	Bagging	Boosting
Training	Parallel	Sequential
Model Dependency	Independent models	Dependent models
Focus	Reduces variance	Reduces bias
Weighting	Equal voting	Weighted voting
Overfitting	Less risk	Higher risk if not tuned
Example	Random Forest	AdaBoost, XGBoost

When to Use

Bagging

- When data is noisy
- When model overfits
- When you want stable results

Boosting

- When you need high accuracy
- When model is weak
- When you can afford more computation

Summary

- Bagging reduces **variance**
- Boosting reduces **bias**
- Bagging works in **parallel**
- Boosting works in **sequence**
- Bagging uses **equal voting**
- Boosting uses **weighted learning**